Rooftop Solar CRM & Project Control Tower

An end-to-end digital operating platform built specifically for NIRT to manage residential and commercial rooftop solar projects: from lead intake and site survey to 80% upfront advance collection, TANGEDCO net metering, and PM Surya Ghar subsidy disbursement.

1. CORE SYSTEM OBJECTIVES

1. ONE CUSTOMER - ONE PROJECT ID

Unified digital record for every client. All EB bills, survey photos, signed contracts, and receipts are tagged to a permanent Project ID (e.g., NIRT-2611-0042).

2. COMMERCIAL CASHFLOW PROTECTION

Enforces NIRT's mandatory **80% upfront advance payment** upon site visit confirmation before factory procurement can be initiated.

3. AUTOMATED WHATSAPP & EMAIL ENGINE

Keeps the customer updated automatically at 8 crucial milestones (site visit date, payment receipt, material dispatch, net metering, subsidy).

4. END-TO-END CONTROL TOWER

A single pane of glass for management to track SLA delays across all 8 departments: Sales, Survey, Liaison, Purchase, EPC, Accounts, and Subsidy.

2. COMMERCIAL PAYMENT TERMS (NIRT OPERATIONAL POLICY)

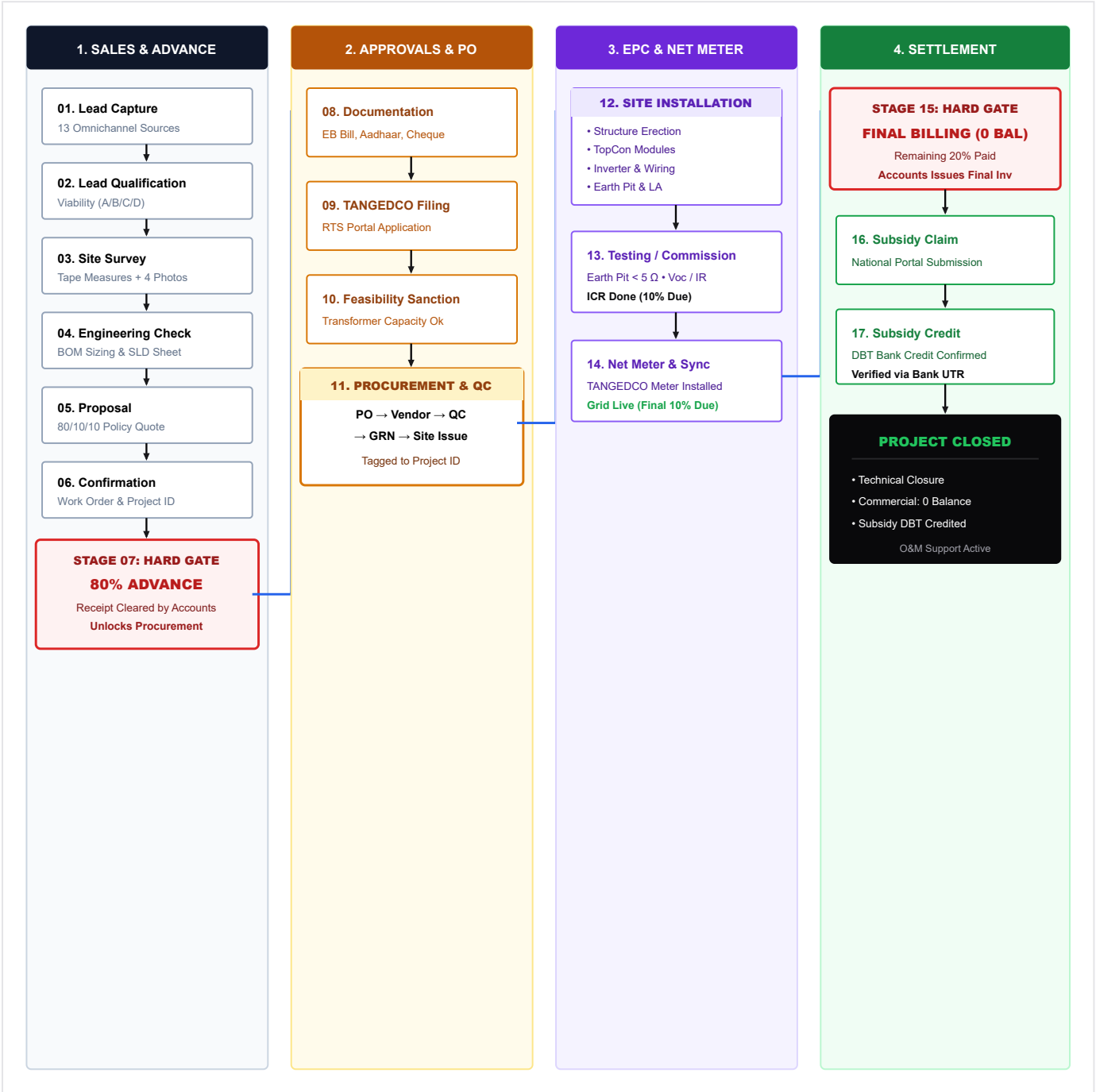
80% ADVANCE	**Initial Payment on Site Visit & Booking:** Collected upon site survey and commercial feasibility sign-off. Secures immediate factory BOM procurement without working capital lockup.
10% INSTALLATION	**Site Work Completion:** Released upon mechanical structure erection, panel mounting, inverter wiring, and earth pit resistance sign-off (< 5 Ω).
10% COMMISSIONING	**Net Metering & Handover:** Final settlement upon TANGEDCO bi-directional meter installation and grid synchronization.

3. DEPARTMENTAL RESPONSIBILITY MATRIX

DEPARTMENT	OPERATIONAL SCOPE	KEY DELIVERABLE
Sales	Lead capture, customer qualification, commercial proposal.	Qualified Deal • Signed Work Order
Engineering & Survey	Roof measurements, shadow analysis, 4 photo slots, SLD upload.	Site Survey Dossier • BOM Sizing
Accounts	80% advance payment receipt, tax invoices, balance zero settlement.	Advance Clearance • Final Tax Invoice
Liaison (DISCOM)	TANGEDCO portal application, feasibility, meter inspection.	Feasibility Approval • Net Meter Sync
Purchase & Store	Vendor PO issuance, material QC inspection, GRN inwarding.	PO • GRN Slip • Site Dispatch Pass
EPC Installation	Structure mounting, DC/AC cabling, testing (< 5 Ω earth pit).	Installation Report • Commissioning ICR
Subsidy Desk	PM Surya Ghar National Portal claim, verification, DBT credit.	Portal Sanction • Bank UTR Record

4. END-TO-END 17-STAGE PROJECT WORKFLOW

Standard Operating Procedure (SOP) governing every installation from first enquiry to final subsidy credit:



5. AUTOMATED WHATSAPP & EMAIL CUSTOMER TRIGGERS

Personalized, bilingual (English & Tamil) alerts automatically triggered to the customer at key transition gates:

MILESTONE TRIGGER	CHANNEL	CUSTOMER NOTIFICATION PAYLOAD
1. Lead Welcome	WhatsApp + Email	Formal greetings from NIRT, executive credentials, assigned solar consultant contact card, and informational solar brochure.
2. Site Visit Confirmed	WhatsApp + Email	Visit date, time window, assigned site surveyor name, mobile number, and roof/EB meter checklist.
3. 80% Payment Done	WhatsApp + Email	Official payment receipt, GST invoice copy, and unified **Project ID** (e.g., NIRT-2611-0042).
4. PO Placed (Factory)	WhatsApp + Email	Confirmation that Tier-1 TopCon panels, inverter, and GI mounting structures are ordered from certified factories.
5. Material Dispatched	WhatsApp + Email	Alert confirming materials passed warehouse QC audit and are loaded in delivery vehicle with expected arrival time.
6. Installation Mobilised	WhatsApp + Email	Site team arrival notice, safety briefing, and daily erection milestone tracking.
7. Grid Synchronized	WhatsApp + Email	Congratulations on TANGEDCO bi-directional net meter synchronization - Solar power generation active!
8. Subsidy Credited (DBT)	WhatsApp + Email	National Portal subsidy approval confirmed with bank direct credit and transaction UTR reference.

6. CORE WEB SOFTWARE CAPABILITIES DELIVERED

- **Lead Deduplication Engine:** Real-time duplicate check on 10-digit mobile and TANGEDCO EB numbers.
- **Site Feasibility Audit:** Tape measurements, shadow analysis, earthing pit space, and 4 photo slots.
- **Triple Pipeline Board:** Visual Kanban tracks (Sales • EPC Execution • PM Surya Ghar Subsidy) with gatekeeper checklists.
- **Pricing & Proposal Sizer:** Instant capacity calculation, 80/10/10 payment breakdown, and official NIRT PDF quotation.
- **360° Project Dossier:** Net metering 8-step tracker, milestone payment ledger, and secure digital document locker.